Milestone 0: Registration

Team Members:

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Track A:

RAG Pipeline

Data Source:

College class notes and assignments

Problem Statement and Idea:

Our project builds an adaptive study assistant for university students who struggle to extract exam-ready, level-appropriate answers from their own course materials — lecture notes, assignment sheets, and handouts. Rather than receiving generic responses that ignore their actual syllabus. The system ingests student-uploaded PDFs, chunks and embeds them using “sentence-transformers” library, and stores vectors in “pgvector” on PostgreSQL, alongside a relational schema (minimum 4 tables, 3NF) tracking document metadata, text chunks with page references, student query logs, and LLM-generated solution sets linked back to their source chunks. A Python interface accepts a natural-language question and a chosen difficulty level (foundational, intermediate, or advanced) retrieves the most relevant chunks via cosine similarity and prompts an LLM (which runs offline) to return a cited, level-appropriate answer.